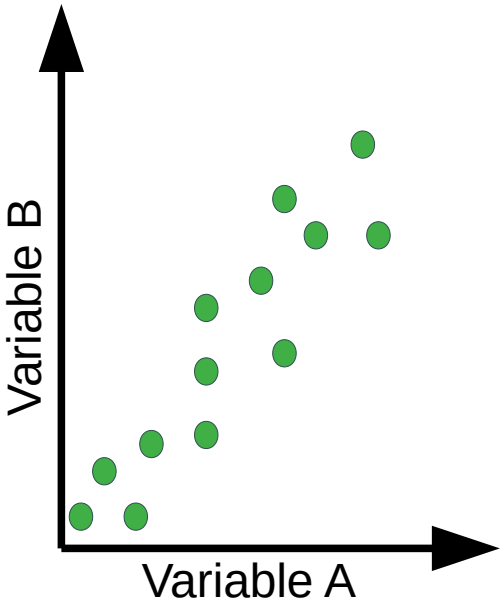
Data analysis and bio-statistics

Practice 5: Variables Relationship

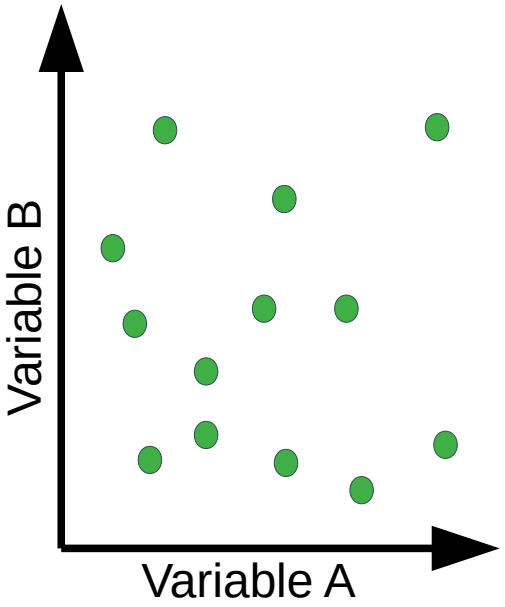
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Pearson Correlation

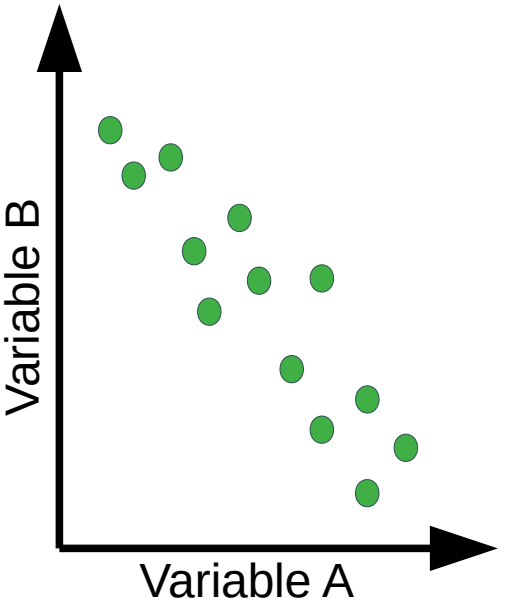
Pearson Correlation Coefficient: A number between -1 and 1



Positive Correlation
Pearson coefficient >> 0
A ↑ → B ↑



Uncorrelated Variables
Pearson coefficient ~ 0
A independent of B



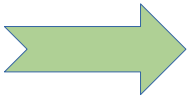
Negative Correlation
Pearson coefficient << 0
A ↑ → B ↓

Correlation Matrix

A tool to visualize the relationship between all variables in a dataset

Dataset

Var A	Var B	Var C	Var D
Observations (Var A)	Observations (Var B)	Observations (Var C)	Observations (Var D)



CorAA = correlation(A,A)
CorAB = correlation(A,B)
CorAC = correlation(A,C)
CorAD = correlation(A,D)

CorBA = correlation(B,A)
CorBB = correlation(B,B)
CorBC = correlation(B,C)
...



Correlation Matrix

	Var A	Var B	Var C	Var D
Var A	CorAA	CorBA	CorCA	CorDA
Var B	CorAB	CorBB	CorCB	CorDB
Var C	CorAC	CorBC	CorCC	CorDC
Var D	CorAD	CorBD	CorCD	CorDD

Correlation Matrix

Correlation Matrix

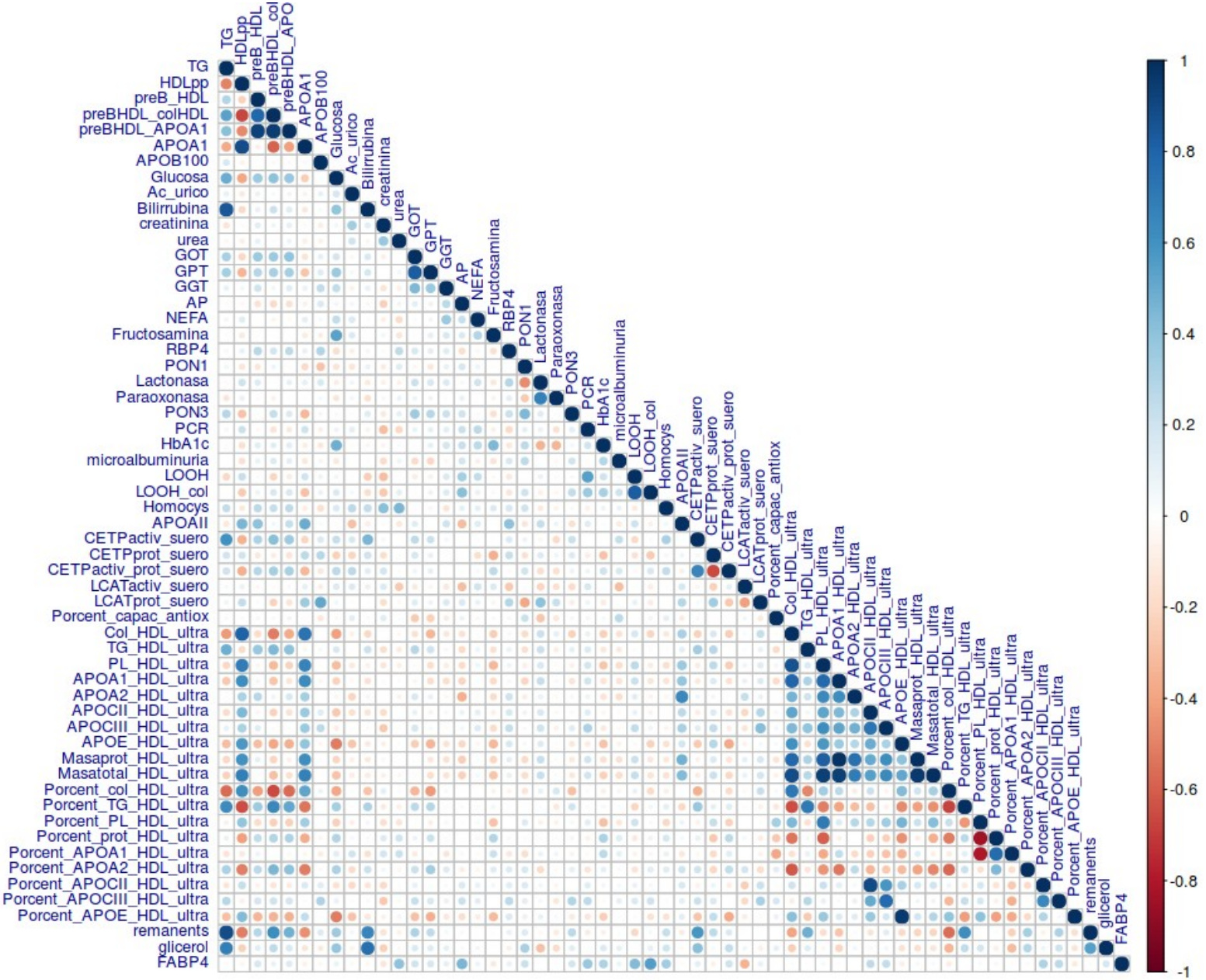
	Var A	Var B	Var C	Var D
Var A	CorAA	CorBA	CorCA	CorDA
Var B	CorAB	CorBB	CorCB	CorDB
Var C	CorAC	CorBC	CorCC	CorDC
Var D	CorAD	CorBD	CorCD	CorDD



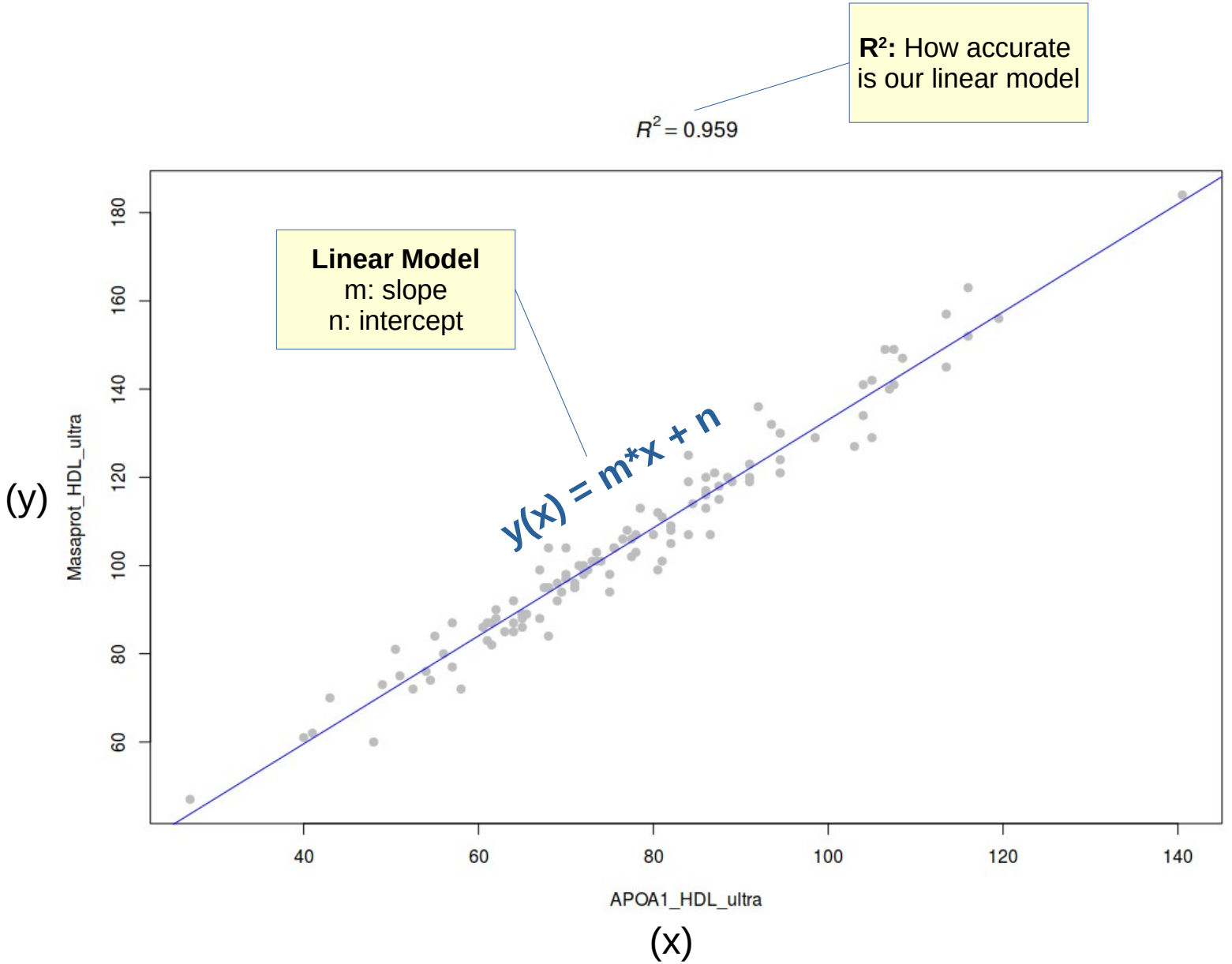
Simplified Correlation Matrix

	Var A	Var B	Var C	Var D
Var A	1			
Var B	CorAB	1		
Var C	CorAC	CorBC	1	
Var D	CorAD	CorBD	CorCD	1

Correlation Matrix Example



Paired Variables Linear Relationship



Linear Model Validation

